

4. Tuple

1. introduction to tuples, immutability.

➤ What is a Tuple?

- Tuple is collection data type In python we can store Multiple values in a single variable.
- Denoted by ()
- Immutable
- Ex:

```
tuple = (10, 20, 30)
print(tuple)
# Output:
(10, 20, 30)
```

➤ What is Immutability?

- Immutability means unchangeable.
- Once a tuple is created, you cannot:
 - Add element
 - Remove elements
 - Modify elements

2.Creating and accessing elements in a tuple.

➤ Creating a Tuple

- A tuple is created using round brackets ().
- For a single-item tuple, you must add a comma after the item.
- It can contain any number of items.
- Ex:
- `tuple = ('hello', 3.14, True, 10)`

➤ Accessing Elements of a Tuple

- You can access elements using indexing.
- Indexing in Python starts from 0.
- You can also use negative indexing to access elements from the end.
- Ex:
- `fruits = ('apple', 'banana', 'mango')`
- `print(fruits[0])`
- Output: 'apple' (first element)

3. Basic operations with tuples: concatenation, repetition, membership.

➤ Concatenation

- You can join two or more tuples using the + operator.
- This creates a new tuple containing elements from all.
- Ex:
tuple1 = (1, 2, 3)
tuple2 = (4, 5)
result = tuple1 + tuple2
print(result).
Output: (1, 2, 3, 4, 5)

➤ Repetition

- You can repeat a tuple multiple times using the * operator.
- Useful for creating repeated patterns.
- Ex:
t = (10, 20)
result = t * 3
print(result)
Output: (10, 20, 10, 20, 10, 20)

➤ Membership

- Use the in or not in keyword to check if an item exists in a tuple.
- Returns True or False.
- Ex:

```
colors = ('red', 'green', 'blue')
```

```
print('green' in colors)
```

```
# Output: True
```

```
print('yellow' not in colors)
```

```
# Output: True
```